CREATING WEBSITES WITH AI

A Model Neutral Beginner's Guide

Jens Jacobsen



A **Chapman & Hall** Book

CRC Press
Taylor & Francis Group

Creating Websites with AI

Anyone can use AI to put a website online in minutes. But a site built in minutes is just any site – not one aligned to your goals, your visitors' needs or the requirements of search engines and AI platforms. The key is to use AI to get better results, not just faster results.

This book provides comprehensive, practical guidance for beginners on leveraging AI to develop websites. Through engaging conversations with AI tools like ChatGPT, Gemini and Claude, readers will learn how to create websites efficiently and effectively. By remaining tool-agnostic, the content is adaptable to suit individual preferences and needs.

Covering every essential aspect – from initial conception to marketing strategies – it offers step-by-step instructions and numerous examples for immediate implementation. It introduces the principle of human-centred development, beginning with fundamental questions of website design, such as defining the goals of the website, understanding the target audience and aligning with their needs. The book then guides readers in selecting the right tool for realising the site, focusing on accessible platforms like Wix, Squarespace, WordPress and Webflow. With AI assistance, the book demonstrates how to create content, add it to the chosen platform and refine the results through feedback from humans and AI alike. It also explores how SEO and marketing strategies can be employed to optimise website visibility on search engines and AI platforms.

Whether the goal is to design a site for a retail store, a medical practice, a restaurant, a personal portfolio or an organisation, this book equips readers with the tools and strategies to confidently develop websites tailored to their unique needs and objectives.

Jens Jacobsen is a highly experienced author of practical, beginner-friendly guides and professional handbooks on web development, user experience (UX) and e-commerce. Across all his work, he is driven by one question: how to convey content in a way that truly resonates with readers. Beyond his books, he shares his knowledge in training courses and seminars and designs all kinds of websites, interactive applications for museums, apps and educational games.



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CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

A CHAPMAN & HALL BOOK

Designed cover image: Shutterstock

First edition published 2027
by CRC Press
2385 NW Executive Center Drive, Suite 320, Boca Raton FL 33431

and by CRC Press
4 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

CRC Press is an imprint of Taylor & Francis Group, LLC

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First published in the German language under the title "Websites entwickeln mit KI"
(ISBN: 978-3-367-10643-1) by Rheinwerk Verlag GmbH, Bonn, Germany.

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ISBN: 978-1-041-21331-4 (hbk)
ISBN: 978-1-041-21330-7 (pbk)
ISBN: 978-1-003-72670-8 (ebk)

DOI: 10.1201/9781003726708

Typeset in Minion
by KnowledgeWorks Global Ltd.

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AI Usage Statement

DeepL Pro was used to generate a draft translation from the original edition. In addition, various AI systems were used throughout the book. These are identified at the points where they are used. The text has been revised throughout by the author. Copyright © 2027 Jens Jacobsen.



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CHAPTER 1

Introduction

Your Guide to Website Success

> From the initial vision to the finished website – let's embark on this exciting journey together. With new AI tools and proven strategies, I will guide you step by step to a successful online presence.

Creating images, texts and even entire websites with artificial intelligence (AI) used to be a distant dream. But today, we have arrived at that very future. Perhaps you have been watching in amazement for several years as AI image generation has become increasingly sophisticated. Or maybe you have played around with early versions of AI chatbots, which could be very entertaining. But even if you are still hesitant about the new technology, AI has been part of our everyday lives for several months now. AI is a tool that sits right in front of us on our desks, waiting to be used. It is currently the dominant technology topic – and rightly so. With AI, we can all now do things that until recently would have taken a very long time or required a team of professionals from a wide range of fields. In this book, I'll show you how to use AI to create a website that's better than any you've ever made. And if this is your very first website, you'll learn how to create a site that will impress even seasoned professionals.

1.1 WHY A BOOK ON THIS TOPIC?

Sure, you can get started right away and type in your website request into ChatGPT, Claude or another AI – and you'll get one right away. The catch: it's just a completely generic website. The AI doesn't ask any questions; it just fulfils your request directly. This approach is fine for playing around. But if you want a website that makes an impact, that people visit, that conveys information or sells products, then you should take a more planned approach. And that's exactly where this book comes in.

DOI: 10.1201/9781003726708-1

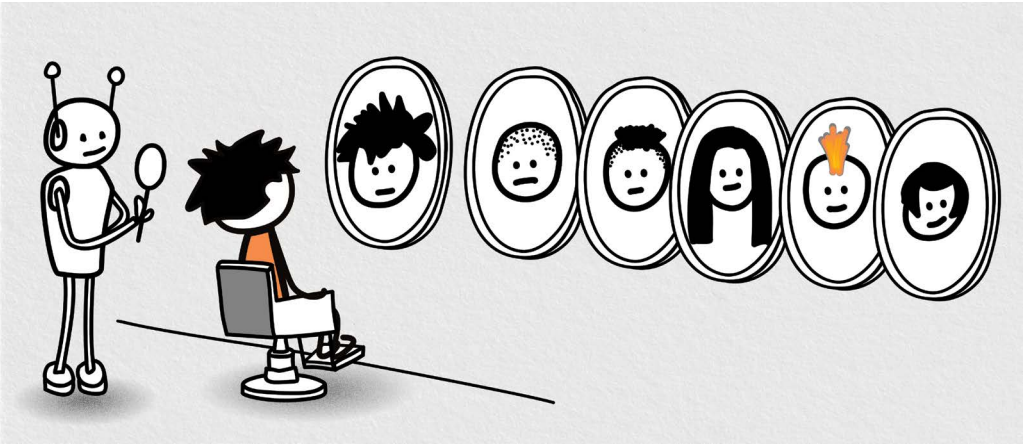


FIGURE 1.1 You need to give clear instructions. "A haircut, please" can produce a wide variety of results – the AI usually doesn't ask questions; it just gets started right away.

1.2 HOW DOES THIS BENEFIT ME?

In this book, you will learn how to systematically plan, implement and make your website successful online using a simple seven-step system. It doesn't matter whether you want a simple *one-page* site that only lists the address and opening hours of your shop, law firm or practice, or a complex website with lots of subpages, masses of content, member areas and many functions. Or a web shop where you want to sell your products and/or services.

Even if you want to create a quick prototype for usability testing or need a website for your start-up, the basic approach is the same. It doesn't matter whether you work alone or in a team or set up the site for other people.

If you only want AI help for certain areas, you'll find everything you need here. Want to develop ideas and create concepts? Or have AI generate text and images? Programme HTML, CSS and JavaScript? Perform search engine optimisation or code optimisation? Do social media marketing or write blog posts? All of this is easily possible, as you'll see.

In addition to well-known AI tools such as ChatGPT, Claude, Gemini and Midjourney, there are dozens more that are very, very useful for creating websites. And there are thousands more that look promising at first glance – but then often prove to be of little help in practice. So that you don't have to try them all out yourself, I'll give you tips on the best tools and their capabilities, which I continuously update online.

TIPS FOR SELECTING THE MOST USEFUL LANGUAGE MODELS

Language models are crucial for creating websites with AI, as you will see repeatedly throughout the rest of this book. That's why I'm showing you a video of the providers and versions that I currently consider to be the most recommendable. Often, these are the

services that you read about most frequently anyway. Sometimes, however, there are a few new AIs that are definitely worth a closer look.

Scan the QR code with your smartphone or enter the URL below into your browser's address bar to see the current overview.

https://bnfr.de/ai093

Or you can ask your favourite AI directly. You can phrase the question as follows, for example:

EXAMPLE PROMPT: OVERVIEW OF AI TOOLS FOR WEBSITE CREATION

I want to create a small website to publish information about my business. To do this, I am looking for an overview of AI tools that can help me with the implementation.

Create a clear list of AI tools that cover the following tasks: conception, wireframes, sitemap, content planning, copywriting, image creation, SEO optimisation, colour scheme creation and design suggestions, as well as responsive design. The tools should preferably be free or inexpensive.

The overview should be presented as a simple list that briefly describes the tools and indicates which of the tasks mentioned they can cover.

The list could look like this:

Tool name: Brief description | Areas of application | Price (e.g. free or price range).

Act as an expert in AI-supported web development who knows all the currently relevant tools and describes them in an easy-to-understand way.

The prompt to copy: bnfr.de/ai085

You will see a URL like this after each sample prompt in the book. Enter it into your browser's address bar and you will be taken directly to a page where you can copy the code with a click to use it in the AI of your choice.

The prompt just shown gives you a good idea of possible tools – but you should treat the AI's recommendations with caution. This is because the AI has also been trained with the providers' marketing texts and countless blogs that briefly describe and positively discuss current tools. In the book, on the other hand, you will only find recommendations that have been thoroughly tested on sample projects and are tailored to your specific task.

1.3 WHAT YOU DON'T NEED TO KNOW

Do you have prior knowledge of design, concept, user experience, code or content writing? Great! Then you can perhaps skim through some sections and jump straight to the practical implementation. However, I still recommend that you at least look at each step as I show it here in the book. This is because you probably have a slightly different approach and can see whether you can get inspiration to do things not only faster but also better in the future with the help of AI.

But if you haven't had anything to do with all these areas before, AI will enable you to keep up with the professionals. In the book, you'll find exactly as much background information as you need to create your own personal website. The great thing about AI is that it tirelessly supports us, answering even the most basic questions in a very motivating and helpful way – and at exactly the level of detail we need. If not on the first attempt, then certainly on request.

1.4 WHY CREATE WEBSITES AT ALL?

You may be wondering what role your own website will play in the future. Many people spend several hours a day on social media – whereas the amount of time spent visiting websites has been declining for years. Large language models and AI assistants are already searching the web for specific information for us and preparing it in a quick and easy-to-read format – all without us having to deal with dubious sites, click away annoying banners or separate content from advertising. More and more people are therefore no longer searching with Google or Bing, but with ChatGPT or Le Chat. In the medium term, such AIs could become our central point of contact for information, shopping and entertainment. We would then hardly ever see independent websites. AI shopping assistants search for products that are a perfect fit for us, find the shop with the best price and place the order without us ever having to visit an online shop ourselves.

And yet your own website is and remains the only place online that you have complete control over. Social media platforms frequently change algorithms and terms of use; they come into fashion and become obsolete. AI providers operate closed systems, where you have no insight and cannot control them in any way. If you use third-party services, you have to play by their rules. And the providers who process your content first have to find it. Your website is indispensable for this. Just as we have had to make our content findable for search engines for decades, we now also have to make our content accessible to AI.

When searching for information beyond short videos and text snippets, a central, reliable source is important. On your website, you present your services, products, references, values and standards – just as *you* want to present them. Even if AI tools increasingly collect and process content, sources are evaluated, and high-quality, well-structured websites can score points here in the long term. Your own website is the means to digital sovereignty. It allows you to define your brand and convey your credibility. A

good site conveys professionalism and builds trust. A website remains indispensable, especially for detailed information, storytelling, branding and serious communication with customers.

1.5 BUILDING WEBSITES FOR PEOPLE WITH AI

Who decides whether your website is a success? The people who use it. People need to visit your site, take in the information you provide (directly or via AI), contact you, book your services or buy your products. This means you need to engage with the people for whom you are developing the site. That's why today's really good websites (and apps and products in general) are created using ***human-centred development***. To do this, you need to take a detailed look at everyone you want to reach with your site – your target groups. And AI is a great help here – the large language models have been trained with vast amounts of data about people, have seen millions of websites and can thus access an immeasurable treasure trove of information about what people want and do on the web.

This is another reason why AI is so helpful in planning the ***user experience*** (***UX***). User experience encompasses all aspects of a person's interaction with a product – from their expectations to their feelings during use and to their experiences afterwards. ***Usability***, i.e. ***user-friendliness***, is an important part of this and describes how easily and effectively people can use a product. With their comprehensive databases, AIs can help you identify and avoid usability problems early on – they can draw on insights from

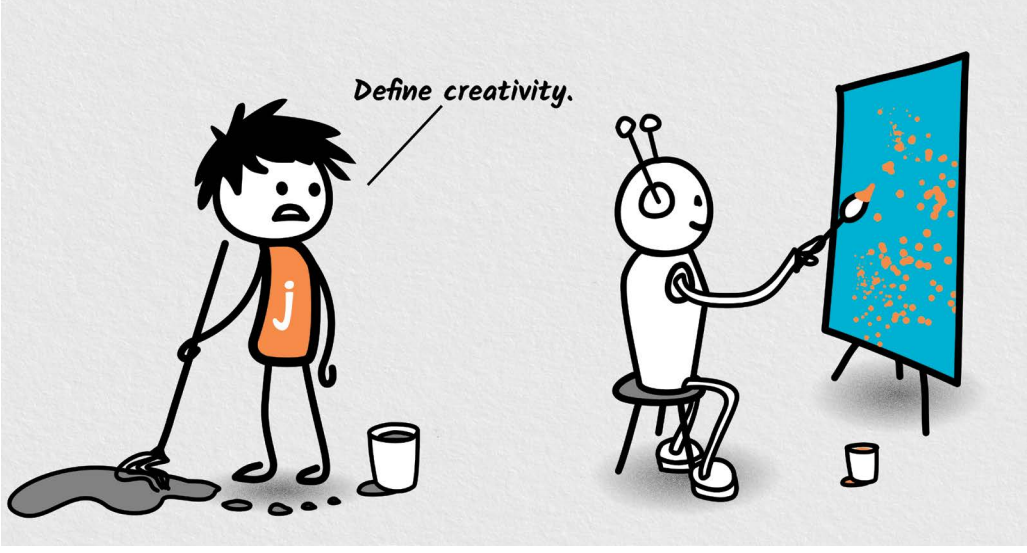


FIGURE 1.2 The advent of AI has fuelled the debate about what creativity is. If we only look at the result, it is becoming increasingly difficult to distinguish between AI-generated and human-made content.

countless website analyses and user studies. And they can help involve real people in the development process, which is still essential if a site is to be successful with users.

1.6 SEVEN STEPS TO SUCCESS

Figure 1.3 shows how your site is created step by step in this book with the help of AI. The graphic also shows the structure for the book's layout. The area in the middle is very important in terms of human-centred development: **gather feedback**. Ideally, you should have people check repeatedly whether what you have developed so far is understandable, useful and appealing to them.

You will learn this in the individual chapters:

Introduction: Your Guide to Website Success

This is the chapter you are currently reading. After this overview of the chapters in the book, you will learn which sample projects you can look forward to and how much effort and budget you should allocate for projects of this size.

Step 0: What AI Can Do and What You Should Know about It

In Chapter 2, I will teach you the necessary basics without burdening you with technical details. You will learn how AI is used in the areas of text, code and image

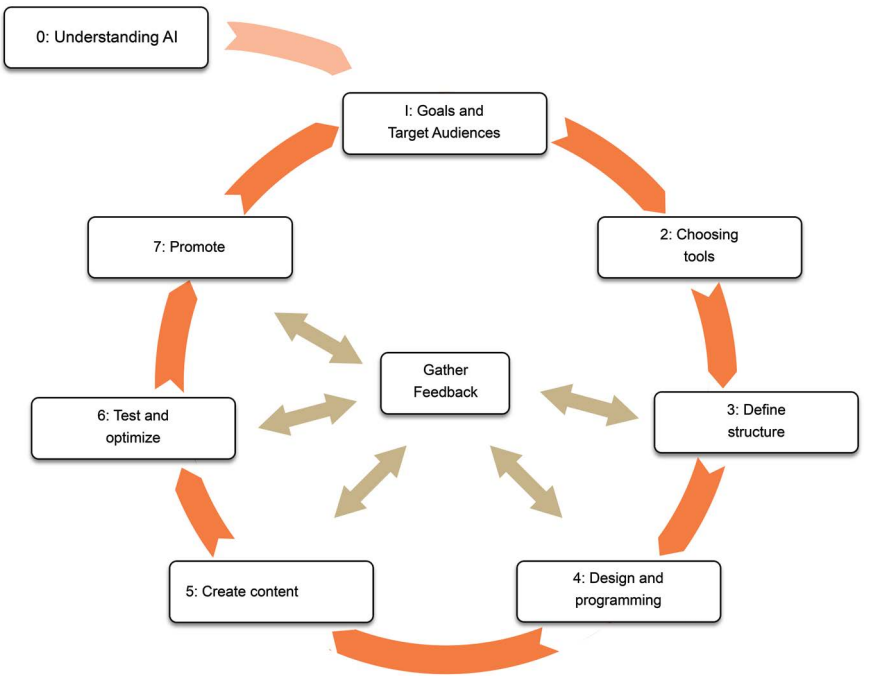


FIGURE 1.3 Develop a successful website in seven steps.

generation and what possibilities and limitations it has in each case. And you will get valuable tips on how to use AI more efficiently and enjoyably.

Step 1: Plan the Route – Identifying Goals and Target Groups

Before you start implementing your website, define clear goals and get to know your target audience inside out. In Chapter 3, you can read about the questions you should ask yourself and how to find the answers with the help of an interactive AI design assistant. You will create a well thought out concept that forms a solid foundation for your website.

Step 2: Select the Best Tools

In Chapter 4, you will discover the best tools and platforms for implementing your site. Learn how to use AI to select the optimal tools and what criteria are crucial in this process. I will give you an overview of website builders, low-code platforms and writing your own HTML/CSS. This will help you find the tools that best suit your project and your prior knowledge.

Step 3: Determine the Optimal Structure of the Website

A good website structure is the key to a positive user experience. Chapter 5 shows you how to create sitemaps, scribbles and wireframes using simple tools. Learn how to use AI to optimise the overall structure of your website as well as the structure of individual pages and to present content clearly.

Step 4: Design Meets Code – Designing and Programming the Site

Chapter 6 is about designing and implementing the basic design of your website. You will learn how to create graphics and templates and implement the individual pages of your website. I will show you how to use AI tools for design creation, how to create pages in HTML or in a construction kit, and which tools are particularly recommended.

Step 5: Efficiently Create Content That Inspires

Content is the heart of your website. Chapter 7 shows you how to create appealing texts and images with AI tools. Examples of effective prompts help you achieve good results quickly. I explain how to integrate content into your website in a meaningful way and which legal requirements, such as imprint and privacy policy, you need to observe.

Step 6: Test and Optimise

In Chapter 8, you will learn how to continuously test and optimise your website. Discover different methods for improving usability and see how AI can help you do this. I will also show you the limitations of AI testing and how usability testing can fill these gaps. Optimise your website's performance and use the checklist to exploit its full potential.

Step 7: Promote and Distribute the Site

A great website is of little use if no one knows about it. In the final step, you will learn how to effectively promote your website. Find out how to use AI for marketing strategies and what dangers lurk there. I will show you techniques for using a wide variety of social media and the possible ways in which AI can help you there. You will also learn how search engine optimisation works in the age of AI and how to make your website optimally accessible for search engines and AI.

1.7 OUR SAMPLE PROJECTS

I always want to show you as concretely as possible how to work with the techniques presented. That's why I've come up with five sample projects that you'll encounter repeatedly throughout the book. The projects are intentionally kept rather small because it's easier to grasp the specifics and demonstrate how everything is implemented. But, of course, you can use all the techniques shown for much larger websites.

The hair salon **Haarte Zeiten Hamburg** is a classic barbershop for men that focuses on quick, uncomplicated haircuts at fair prices. Master hairdresser Lars Schneider and his team value a down-to-earth atmosphere without any frills. The website is designed to reflect this approach – it only provides the most important information, such as opening hours, location and a few meaningful photos. It deliberately avoids additional features such as online appointment booking.

Fit-mit-Frieder-und-Fabi are two coaches with degrees in sports science who offer fitness courses ranging from running training to ice skating workshops. In addition to the usual information about courses and training locations, their website also offers a video library that users can subscribe to. The focus is on a motivational web presence that conveys the authentic personalities of the two coaches.

Fellfreunde Fulda e. V. is a non-profit organisation that cares for stray dogs from Greece and Romania and places them with families in Germany. The website serves as a platform to present the work of the association and to find homes for the dogs. With authentic photos and testimonials from adoptive families, the organisation builds trust and demonstrates the impact of its work.

The Oldenburg **Bistro Verde** is a cosy café with healthy and sustainable cuisine, particularly known for its vegan dishes. Owner Sophie Becker has created a place that focuses on relaxation and well-being. The website is designed to present the usual information such as opening hours and menu and show authentic photos of the dishes and the ambience.

Finally, **Paper Love** is a charming boutique print shop run by siblings Sabrina and Tim Weber, who produce handmade greeting cards, art prints and postcards. Their website is intended to be a digital gallery of their work, with each piece presented

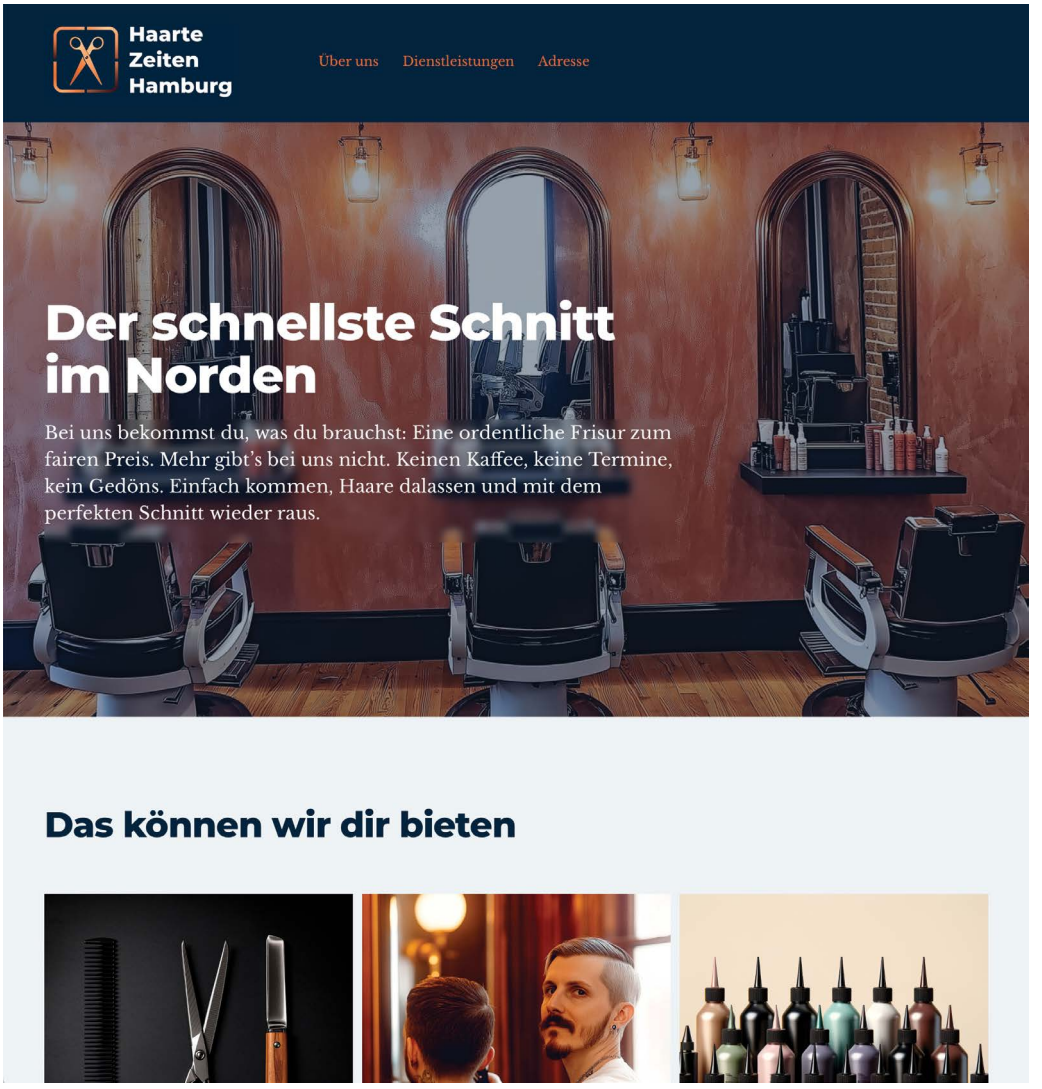


FIGURE 1.4 The website for our example project, the barber shop "Haarte Zeiten Hamburg". (Figure created using Squarespace.)

with a personal story. They have deliberately decided not to have an online shop, focusing instead on direct contact with their customers – they believe that art thrives on personal connection.

1.8 COSTS AND TIME REQUIRED

Let's have the AI estimate how much time and money we would need to invest to implement each of these projects. You can do this with this prompt, which you can enter in ChatGPT, for example:

EXAMPLE PROMPT: COST ESTIMATE WEBSITES

I would like to create a well-founded plan for the implementation of five websites. The aim is to realistically estimate the necessary resources, time and costs in order to implement the projects efficiently.

Analyse the requirements and create a table containing the following information: estimated number of HTML pages per website, estimated number of images, which should preferably be created using AI (and, if necessary, through traditional photo shoots), as well as the effort required for conception, design and implementation in days. Add a total cost estimate in euros.

The end result should be a table in the following format:

Name of site | HTML pages | Total number of AI-generated photos | Total number of real photos | Concept development time [days] | Design time [days] | Programming/implementation time [days] | Costs [€]

The table should be delivered in a precise, clear format and contain practical, comprehensible figures.

Act as an experienced project manager in an agency that implements dozens of small and medium-sized websites every year at very reasonable prices. Use your expertise to provide realistic and cost-effective recommendations.

The prompt to copy: bnfr.de/ai084

This gives us a rough idea of how a small agency would traditionally calculate the cost of creating these sites. The AI took the values it found in its data and estimated where it couldn't find any. It's impossible to say where this is the case. The range of variation in such estimates is enormous, as are the price differences from agency to agency. But based on my experience, the values seem plausible to me as a point of reference, even if they are set quite low. You can also consult different AIs to see other opinions – but in the usual language models, I have received answers that are quite similar to those shown in Figure 1.5.

Name of site	HTML pages	Total number of AI-generated photos	Total number of real photos	Concept development time [days]	Design time [days]	Programming/implementation time [days]	Costs [€]
Haarte Zeiten Hamburg (Barbershop)	5	6	10	2	3	3	2 400
Fit mit Frieder & Fabi (Fitness Coaching)	10	12	8	3	4	5	3 600
Fellfreunde Fulda e.V. (Nonprofit)	12	15	15	4	5	6	5 200
Bistro Verde (Café)	6	8	10	2	3	4	2 800
Paper Love (Print Boutique)	8	10	12	3	4	4	3 400

FIGURE 1.5 ChatGPT's effort estimation. (Figure created using ChatGPT 4o.)

Just don't make the mistake of asking how AI can help us reduce costs or effort. This is because that way we fall into the trap that countless companies have fallen into, wanting to save costs and then going bankrupt. Because when everyone suddenly has a tool at their disposal to achieve significantly better results with the same effort, those who do so will survive in the market. On the other hand, those who only use the tool to save a few pounds in the short term or to be able to go home a little earlier will soon be squeezed out. Because this aspect is so important, more on this in the next chapter.

1.9 IT'S ME, YOUR AUTHOR

The whole book is about you and AI, but you may want to know who is behind this book. So here is my brief story about what connects me to the topic of the web.

Sometimes it takes a coincidence to send you in the right direction. For me, it was an internship in the late 1990s, when no one in the office knew how to change a paragraph of text on the website. With the book *Teach Yourself HTML in 14 Days*, I immersed myself in the world of web design – and was quickly fascinated by the possibilities of the internet. What began as a technical challenge quickly developed further: from pure programming, I moved on to conception: what content and functions does a website



FIGURE 1.6 The man behind the AI book, Jens Jacobsen.

actually need? In my search for systematic approaches, I discovered usability, which was still relatively unknown at the time. Suddenly, it was no longer just about creating websites, but about designing them in such a way that people would enjoy using them. In hundreds of usability tests since then, I have observed people navigating websites, searching for information and often getting frustrated in the same places. These direct insights into user behaviour have taught me what really works – and what doesn't. As the focus shifted from usability to the broader user experience, my toolbox expanded to include new methods and perspectives. I have been sharing these insights for many years in books such as *Website Design* and *The Practical Guide to Usability and UX* (co-authored with Lorena Meyer), as well as in my role as a lecturer at Diploma Hochschule and in countless workshops. One thing has never changed: my enthusiasm for designing digital offerings that help people achieve their goals.

Today, I have come full circle: with the advent of AI tools, I am once again creating HTML code myself – but this time with an ease that I could only dream of back then with my HTML manual. As someone who has actively accompanied the development of the web from the first hand-coded pages to usability and complex UX strategies, I would like to accompany you on your journey into this new era with this book – with practical knowledge from countless projects and a clear focus on what your users – and your websites – really need.

1.10 HOW MUCH AI IS IN THIS BOOK?

This book is not just a guide to using AI – it is itself a product of collaboration between humans and machines. Throughout the entire creation phase, AI was my critical sparring partner: it helped me to question ideas, think concepts through and identify potential weaknesses before they could become real problems. And as soon as I had finished the first draft of a chapter, I submitted it to my self-programmed AI jury: a novice in the field of website creation, an experienced computer book editor and a super-critical sceptic each gave feedback and pointed out areas for improvement. This allowed me to give my human test readers with a request for feedback a much more mature version than before.

The five sample websites you will find in this book are a direct result of my collaboration with AI. From the initial concept sketch to the finished code, I deliberately went through the entire creation process with AI, as shown in the book. I wrote the texts with their help, generated the images with AI and wrote the HTML, CSS and JavaScript code in dialogue with AI systems. This allows you to not only see the end result but also follow the entire process step by step – with all the surprises, challenges and "aha" moments.

It was fascinating to experience how AI often brought unexpected perspectives and inspired me to come up with creative solutions that I might not have found on my own. What began as a methodological approach to the book project developed into a veritable field of experimentation: the more I worked with the various AI tools, the



FIGURE 1.7 Throughout the book, I adhered to the golden rule: treat AI as a partner, not a pack mule. (Image generated with Midjourney v6.1.)

more possibilities I discovered and the more fun the creative process became. These experiences are directly incorporated into the practical tips and instructions you will find in this book.

1.11 THANKS TO

While AI tools open up new possibilities for us every day, it was above all the conversations with people that gave this book its depth and relevance. My special thanks go to:

Viko Mann for the exchange on personas, as well as Michael Überschär and the German UPA's Artificial Intelligence Working Group for the technical discussions

Monika Schirmeier for her insights on legal aspects and sustainability

Wiebke Mülhoff and my students at Diploma Hochschule, whose fresh perspectives have shaped my understanding of the digital generation's approach to AI

Franziska Kaiser for her philosophical explorations of AI, art and design

The readers of my newsletter at benutzerfreun.de for the inspiring exchange via email, blog comments and video conferencing

The team at Rheinwerk Verlag, without whom this book would not exist, especially my editor Erik Lipperts

I would like to express my heartfelt thanks to my wife Cornelia – not only for her support during the many hours of writing but also for her infectious enthusiasm and willingness to experiment with AI and for the valuable exchange of ideas.

And, of course, I would also like to thank the developers of the AI tools, whose digital creations made this book possible and who, between moments of amazement and head-shaking, taught me time and again how machines and humans think.